

Figura 5. Amplificador en fuente común con desacoplo de fuente.

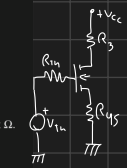
Especificaciones:

- **Resistencias**: R1 = 30 k Ω; R2 = 10 k Ω; R3 = 1 k Ω; R4 = 400 Ω; R5 = 100 Ω; RL = 1 k Ω.
- **Condensadores:** C1 = C2 = 10 µC; C3 = 220 µC.
- **Alimentación:** V_{DD} = 12 V.
- **MOSFET 2N7000**: $K = 0.128 \frac{A}{V^2}$; $V_T = 2.1\ V$.

N. ideal porque, pero

$$\Rightarrow V_{TH} = \frac{R_2}{R_1+R_2} \cdot V_{CC} = 3V$$

$$R_{TH} = R_2 // R_1 = 7{,}5K\Omega$$



¿Corte?



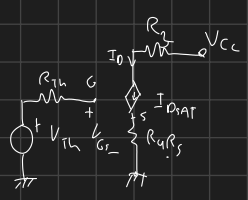
$V_{GS} = V_{TH} = 3V$

$V_{GS} \leq V_T$ X

$3 \leq 2{,}1$ X

NO CORTE

¿Saturación?



$$V_{GS} = -I_G \cdot R_{TH} + V_{TH} - R_{dS} \cdot I_{DSAT} =$$

$$= V_{TH} - R_{dS} \cdot \frac{K}{2}\left(V_{GS} - V_T\right)^2 =$$

$$= V_{TH} - \left(\frac{R_{dS} \cdot K}{2}\right)\left(V_{GS}^2 + V_T^2 - 2V_{GS}V_T\right)$$

$$\Rightarrow V_{GS} = V_{TH} - \left(\frac{R_{dS}K}{2}\right)V_{GS}^2 - \left(\frac{R_{dS} \cdot K \cdot V_T^2}{2}\right) + \left(\frac{2R_{dS}K}{2}\right) \cdot V_{GS}V_T$$

$$V_{GS} = 3 - 32V_{GS}^2 - 1{,}44{,}72 + 134{,}4V_{GS}$$

$$32V_{GS}^2 - 133{,}4V_{GS} + 138{,}72 = 0$$

$$V_{GS} = \begin{cases} 1{,}91V > V_T X \\ 2{,}25V > V_T \checkmark \end{cases}$$

$$I_{DSAT} = \frac{K}{2}\left(V_{GS} - V_T\right)^2 = \frac{0{,}128}{2}\left(2{,}25 - 2{,}1\right)^2 = 1{,}44\ mA$$

$$V_{DS} = V_{CC} - I_D\left(R_3 + R_4R_5\right) = 12 - I_D(1500) = 9{,}84V$$

$$V_{DS} \geq V_{GS} - V_T \;//\; 9{,}84 \geq 0{,}15V$$